

3-Month DSA Roadmap (Beginner to Confident)

This roadmap is designed for students who want to build strong DSA skills in 3 months with a clear and simple direction. It is especially helpful for beginners and students from Tier 3 colleges who want to stand out in placements.

Simple rule to follow:

First build strong basics till Linked List and Trees, then move to advanced topics.

This plan is not about rushing. It is about understanding topics step by step and becoming confident in problem solving.

Month 1: Build the Base (Most Important Phase)

- **Week 1: Arrays** – Learn traversal, searching, prefix sum, and basic sliding window. Focus on understanding patterns.
- **Week 2: Strings** – Frequency count, two pointers, palindrome-based problems, and basic string manipulation.
- **Week 3: Recursion + Sorting** – Understand how recursion works. Learn bubble, selection, insertion sort and the idea behind merge sort.
- **Week 4: Linked List** – Insert, delete, reverse a list, find middle node, and understand pointer-based thinking.

Month 2: Core DSA (Where Your Thinking Improves)

- **Week 5: Stack** – Basics, valid parentheses, next greater element, and understanding LIFO logic.
- **Week 6: Queue** – Basics, circular queue idea, and sliding window understanding.
- **Week 7–8: Trees** – Binary tree basics, DFS traversal, BFS traversal, height of tree, and simple diameter logic.
- If you are from a Tier 3 college, focus deeply till this point. Strong basics till Trees can already help you in many interviews.

Month 3: Advanced Level (After Basics Are Strong)

- **Week 9: Binary Search** – Standard problems, rotated array, and answer-based binary search.
- **Week 10: Hashing** – HashMap usage, frequency problems, and subarray sum concepts.
- **Week 11: Heap / Priority Queue** – Min heap, max heap, and top K type problems.
- **Week 12: Graphs (Intro)** – Graph basics, BFS and DFS traversal.

Tips to Do DSA (For Beginners & Mediocre Level Students)

- Always do a dry run before coding. It helps you understand the logic clearly.
- In the beginning, it is normal to not understand questions. Give yourself time.
- You can refer to the solution when you are stuck, but don't depend on it.
- After seeing the solution, close it and write the full code again using your own logic.
- Make sure you do not peek again while rewriting. This is where real learning happens.
- Try to maintain a daily target: at least 5 easy and 2 medium questions.

Revision Plan (Very Important for Real Growth)

The best way to remember DSA is regular revision.

Choose a fixed time slot of 2–3 hours where you only revise previously studied topics.

If you are a beginner:

Solve questions that focus on only one topic at a time (only arrays, only strings, only linked list). This helps in building clarity.

If you are at a medium level:

Start solving mixed questions that include multiple topics together. This improves real problem-solving and interview thinking.

Revision is what makes concepts strong and long-lasting.

Interview Preparation Strategy

- Be very clear with programming basics: loops, functions, conditions, arrays.
- First prepare DSA step by step till Linked List.
- Arrays → Strings → Recursion → Sorting → Linked List
- Once confident till Linked List, then move towards Trees and advanced topics.